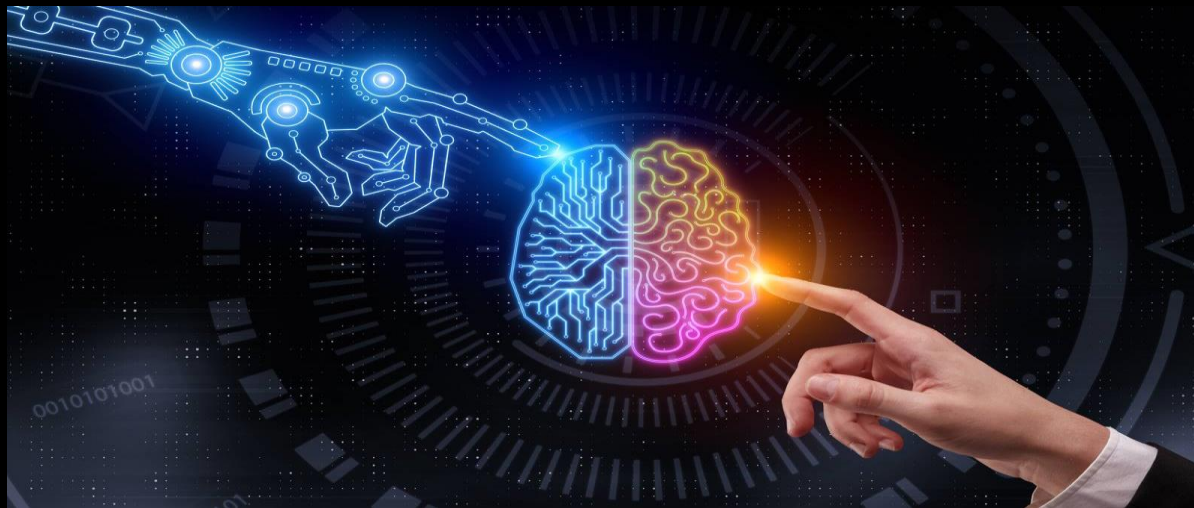


# Modern University for Technology & Information Faculty of Computers & Artificial Intelligence

## The Third Scientific Conference March 2024

الذكاء الاصطناعي : أفاق ومحاذير





**AI 2050**

**AI as LIFE STYLE**

**Dr. Hesham El-Deeb**

**Head of AI Department**

# AGENDA

1. AI Main Categories
2. AI Today (2024)
3. Google World Vision
4. AI Future Vision (2050)
5. Our Daily Life (2050)
6. Frightening AI
7. AI Ethical frameworks

# AI MAIN CATEGORIES

- **ANI** : (**currently**) known as narrow AI or artificial narrow intelligence. Weak AI refers to any AI tool that focuses on doing **one task really well**.  
e.g. Siri by Apple, Alexa by Amazon, Cortana by Microsoft and other virtual assistants, IBM's Watson and Image / facial recognition software.
- **AGI** : (**in progress but not yet**) known as general AI or artificial general intelligence). It exhibits **human-level intelligence**. So, it can understand, think, and act the same way a human might in any given situation (with emotion behavior).  
e.g. Self-driving cars, Speech recognition, like AI chatbots.
- **ASI** : (**Future**) known as artificial superintelligence (ASI) or superintelligence. It is the hypothetical AI that **surpasses human intelligence and ability**. Even the brightest human minds cannot come close to the abilities of super AI.  
e.g. Space exploration: Super AI can facilitate space exploration, as the technical challenges in developing a city on Mars and it is probable that  
**robots may take over the world !!!!!!!**

# AI TODAY

- Today, AI is painting artwork, making music, completely changing how movies are made, preparing food and building recipes for spices,, seeds, and everything that could be done an infinite number of ways.
- **Deep fakes ( AI-generated fake videos )** are lifelike and could easily disrupt world events because **people trust videos so much**.
- **AI-based robots** can read text, charts, and faces/emotions better than humans. Robots are filling all aspects of manufacturing and are beginning to become **companions** to the old, infirmed, lonely, or really anyone.
- At least one robot (**Sophia**) has attained **Saudi citizenship**.
- AI tools using **deep learning** can produce human-like text to produce song lyrics (words), “conversations” with historical figures, translations, text summaries, note conversions and journalistic pieces through complete essays.
- If we think we’ll always be programming AI the way we do today, Google’s new **AI designs are better than humans**.
- **AI is learning how to create itself.**

# PERSONALIZED CHATBOTS

- To get a chatbot in 2024:

- AI giants Google and OpenAI are **gambling big on going small**: both are developing user-friendly platforms that allow people to customize powerful language models and make their own mini chatbots that satisfy to their specific needs—**no coding skills required**.
- Both have launched **web-based tools** that allow anyone to become a **generative-AI app developer**.
- Generative AI models learn the patterns and structure of their input training data and then generate **new data** that has similar characteristics.

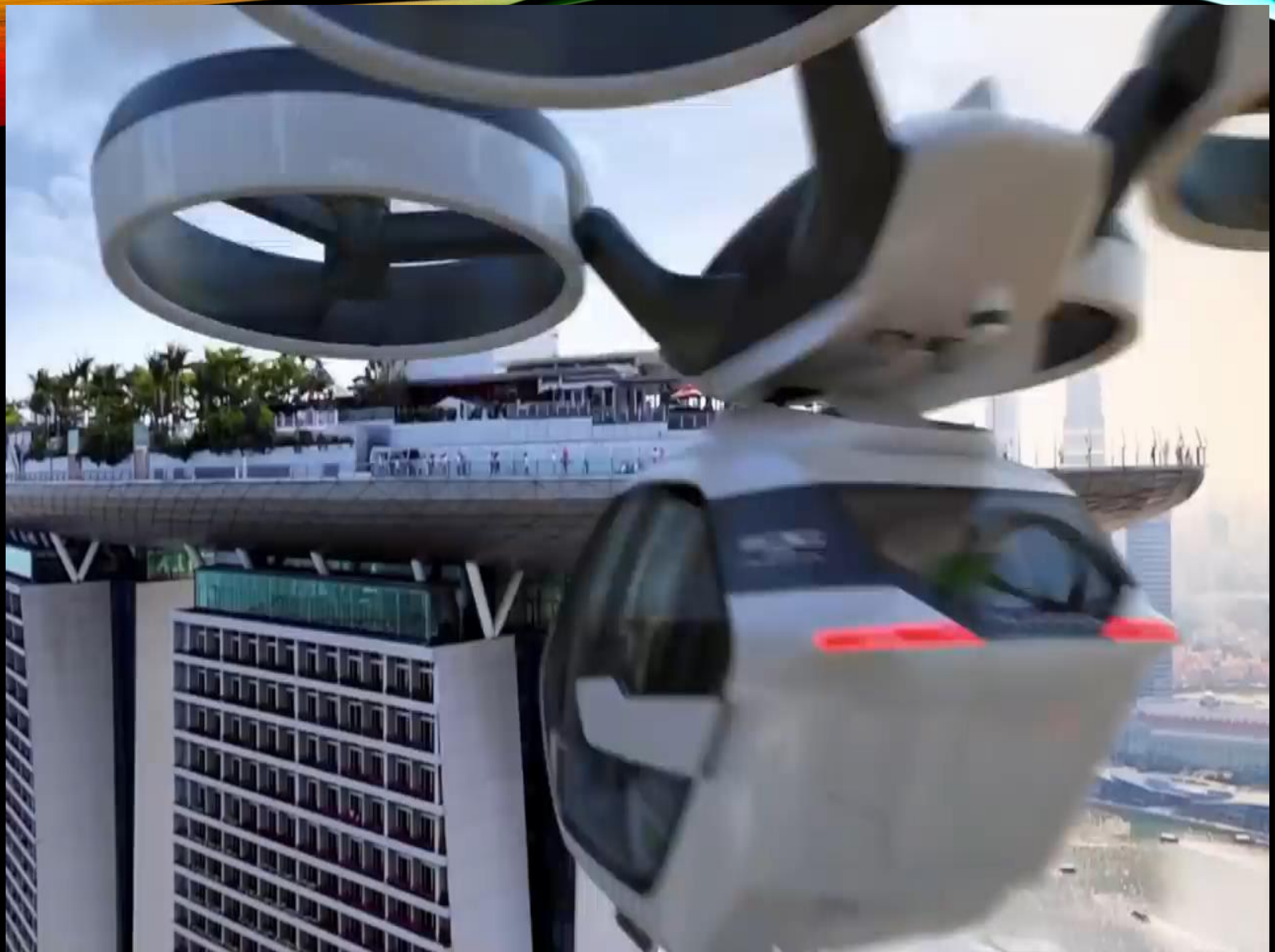
# GOOGLE WORLD VISION

- Google's approach to developing and joining the potential of AI is grounded in its founding mission—to organize the world's information and make it universally accessible and useful—and it is shaped by Google's commitment to improve the lives of as many people as possible.
- **Gemini** developed by Google DeepMind, is a family of multimodal models, announced on December 6, 2023, positioned as a competitor to GPT-4: OpenAI launched on March 14, 2023.
- they can process not only text but images and even videos.

# AI FUTURE VISION (2050)

► There are some **key indicators** of today life that shape that AI future vision in 2050:

1. The Metaverse
2. Entertainment
3. Quantum Computing
4. Automated Transportation Revolution
5. Healthcare Services
6. Workplace Transformation
7. Space Exploration
8. Personal Assistants
9. Education
10. Sport and Fitness



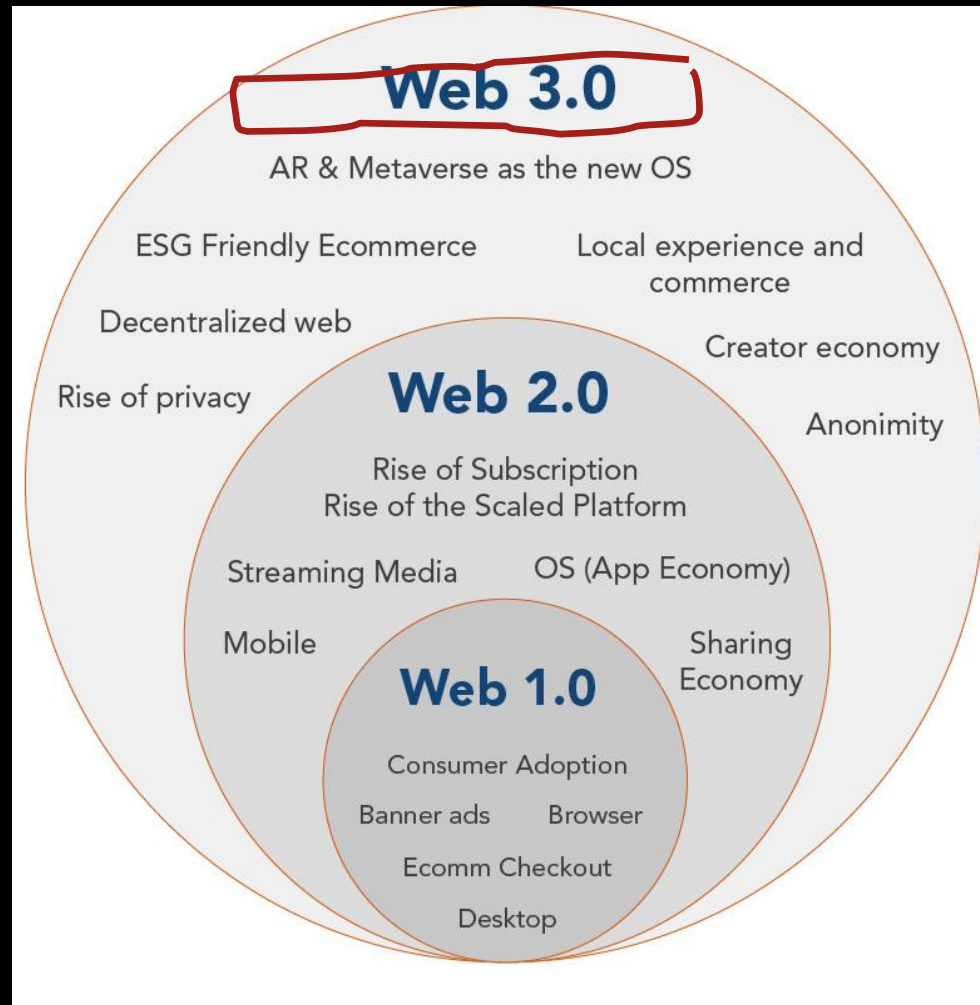
# EVOLUTION OF DECENTRALIZED WEB

The World Wide Web (WWW) refers to all the **public websites or pages** that users can access on their local computers and other devices through the internet.

These pages and documents are interconnected by means of hyperlinks that users click on for information.

This information can be in different formats, including **text, images, audio and video**.

Web 3.0 is an idea for a new iteration of the World Wide Web (WWW) which will be the host of the **Metaverse**.



# 1. THE METAVERSE ( 3D INTERNET)

- Meta stands for **beyond** and verse stands for **universe**.
- It is a virtual worlds in which users represented by avatars interact, usually in 3D and usually focused on **social and economic connection**.
- Metaverse could be defined as a hypothetical iteration of the Internet as a single, universal, and full of virtual world that is facilitated by the use of virtual reality (VR) and augmented reality (AR) headsets.
- In **2019**, the social network company **Facebook** launched a social VR world called Facebook Horizon.
- In **2021**, the company was renamed "Meta Platforms" and its chairman Mark Zuckerberg declared a company commitment to developing a metaverse.



# AVATARS Socializing In The Virtual World *SECOND LIFE*

Proposed applications for metaverse technology include **improving:**

1. work productivity
2. interactive learning environments
3. e-commerce
4. mass-audience interaction
5. healthcare
6. real estate



- In addition to Meta, **tech giants** including Google, Microsoft, Nvidia and Qualcomm are also investing billions of dollars in the concept.
- Management consultancy / McKinsey & Company has confidently predicted that the metaverse economy could reach **\$5 trillion by 2030.**

# THE METAVERSE (2050)

- By 2050, we will be unable to tell the difference between a virtualized real person and an AI-driven avatar. **Avatars will be as universal as the cell phone is today.**
- We will virtually be **able to travel the world and infinite custom planets, all from home.**
- It will be our parallel life, connected via multiple devices, wearables and even our brain. **It will be mixed reality.**
- Many people will select to spend much of their day in virtual worlds where they can become whoever they want -- a “**life**” **with no limits.**
- Everyone could make his **digital twins** from his relatives and friends to chat with them if they die (**immortal best friends**).

## 2. ENTERTAINMENT(7)

1. Most people will have access to artificial general intelligence (AGI): using the virtual reality, quantum computing.
2. AI can make two-hours movies that are photorealistic:  
Actors will be **immortals**, they will not die or stop shining on cinema screen. All we need is their videos, T.V interviews to make a new two-hours movies in one afternoon using their avatars which will last forever. No further using of cameras to produce war scenes by public cinema extras كومبارس السينما. **Hollywood studios will vanish unless they change their investments.**
3. Photorealistic virtual worlds are mostly generated by AGI: using only **one text paragraph** will be enough to generate a **whole AI virtual city or virtual forest**. Movies will be sensitive to **spectaculars emotions** and its ends changes according to their status and modes.

# ENTERTAINMENT

4. Humans will be able to merge to artificial intelligence: by accessing **numerous virtual web sites** having infinite number of photos, videos and they can perform predictions based on its contents.

5. Humans can communicate with each other telepathically: **توارد الخواطر**

The supposed communication of thoughts or ideas.

6. AIs are writing better software in seconds than the human software developers:

AIs software to be processed by quantum computing generating video games or virtual reality games in **80% less time and effort**.

7. AIs are writing better books and screenplays **سيناريوهات** than humans:

AI can write novels. poems and music concretes on creative basis which superimpose humans creativity.

# OpenAI TOOL: **SORA**

( CREATING REALISTIC AND IMAGINATIVE SCENES FROM TEXT INSTRUCTIONS)



# 3. QUANTUM COMPUTING

## الحسابات الكمية

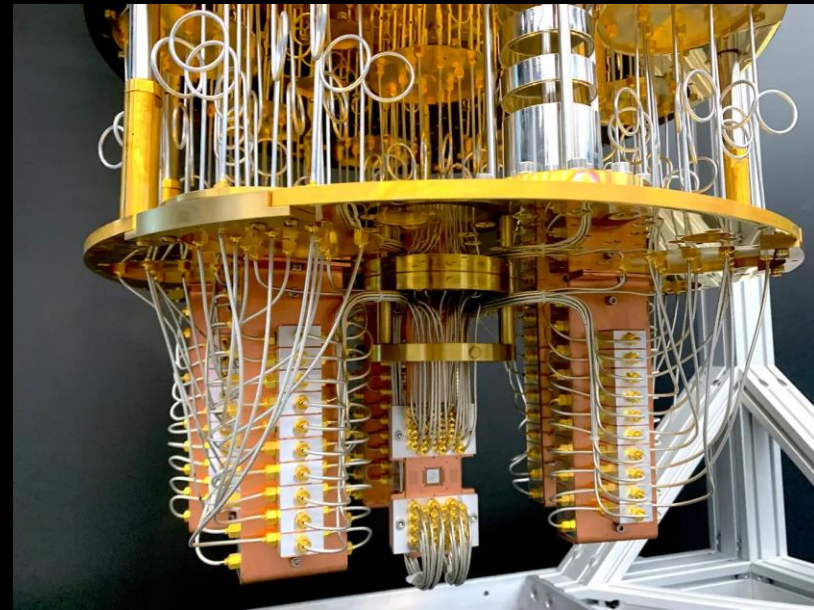
- Quantum computing is a multidisciplinary field comprising aspects of **computer science, physics, and mathematics** that utilizes quantum mechanics to solve complex problems faster than on classical computers ( almost Real time computations ).
- Applications of quantum computers include:
  - machine learning (ML)
  - optimization
  - simulation of physical systems
  - the simulation of chemical systems
  - solving problems that are currently impossible for even the most **powerful supercomputers** on the market.
- Quantum mechanics is the area of **physics** that studies the behavior of particles at a **microscopic level**, with their equations that **describe how particles behave** in the world around us.

# QUANTUM COMPUTING

- We are moving through the era of **bits** and bytes as our computing frame of reference. The future is about quantum computing. Quantum computing uses **qubits**, which can be **any proportion of the 0 to 1 state**.
- qubits, are represented by quantum particles. The manipulation of qubits by control devices is at the core of a quantum computer's processing power.
- In classical computing, a bit is an electronic signal that is either on or off. The value of the classical bit can thus be one (on) or zero (off). However, because the qubit is based on the laws of quantum mechanics it can be placed in a **superposition of states**.
- Early testing has shown a dramatic, exponential speed up in queries. All **calculations happen simultaneously**.
- **Searching large data sets**, which will still be quite interesting in 2050, will only be done with quantum.

# IBM QUANTUM COMPUTER

- Supercomputers services will be on hand using **cloud computing services**.
- Not only for real time computations of vast amount of data (**Big Data**) but also for predication using new Neural networks technology called **Quantum neural networks**.
- deep learning and traditional neural networks will **be old fashion**.



# QUANTUM COMPUTING PRINCIPLES

Quantum principles require a new dictionary of terms that include:

- **Superposition** تراكب (Required)

Superposition states that, much like waves in classical physics, you can add two or more quantum states and the result will be another valid quantum state. Conversely, you can also represent every quantum state as a sum of two or more other distinct states. This superposition of qubits gives quantum computers their **inherent parallelism**, allowing them to process millions of operations **simultaneously**.

- **Entanglement** تشابك (Required)

Occurs when two systems link so closely that knowledge about one gives you immediate knowledge about the other, no matter how far apart they are. Quantum processors can draw conclusions about one particle by measuring another one. Quantum entanglement allows quantum computers **to solve complex problems faster**.

- **Decoherence** فك الترابط (Not Required)

It is the loss of the quantum state in a qubit. **Environmental factors, like radiation or noise can cause the quantum state of the qubits to failure**. A large engineering challenge in constructing a quantum computer is designing the various features that attempt to **delay decoherence** of the state, that **shield the qubits from external fields**.

# HOW CAN AWS HELP WITH QUANTUM COMPUTING?

- Amazon Braket as an Amazon Web Service (AWS) is a fully managed quantum computing service. It is designed to help speed up scientific research and software development for quantum computing.

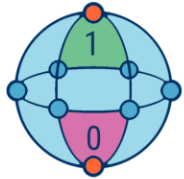
## Amazon Braket do the following tasks:

1. Work with different types of quantum computers and circuit simulators using a consistent set of development tools.
2. Build quantum projects on a trusted cloud with straightforward pricing and management controls for both quantum and classical technology.
3. Innovate quickly with expert guidance and tech support, or collaborate with consultants in the Amazon Quantum Solutions Lab.

# Quantum Computing

Vs.

# Classical Computing



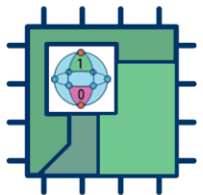
Calculates with qubits, which can represent 0 and 1 at the same time

Calculates with transistors, which can represent either 0 or 1



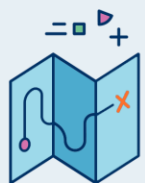
Power increases exponentially in proportion to the number of qubits

Power increases in a 1:1 relationship with the number of transistors



Quantum computers have high error rates and need to be kept ultracold

Classical computers have low error rates and can operate at room temp



Well suited for tasks like optimization problems, data analysis, and simulations

Most everyday processing is best handled by classical computers



- Qubits are essentially quantum transistors.
- Its materials from nanosized crystals .
- Google's qubits are made of aluminum.
- IBM uses a mix of aluminum and niobium ( silver-gray metal of the transition series).

## 4. AUTOMATED TRANSPORTATION REVOLUTION

- The roads, skies, and seas will be busy with autonomous vehicles. From self-driving cars (UGV) to unmanned aerial vehicles (UAV) and autonomous ships (USV), AI will navigate us through a safer, more efficient **world of transportation, reducing accidents and optimizing travel time.**
- The increased population of 2050, though mainly working from home, will still need to get around.
- The **Society of Automotive Engineers** (SAE) defines six levels of vehicle driving automation systems ( **see next slide** ).
- We're now at **level two**, with cars able to control steering, acceleration, and braking, while still requiring steering wheels and drivers to remain engaged.

# Different Cars Are Capable Of Different Levels Of Self-driving On A Scale Of 0-5 ( 6 levels )

- Level 0: All major systems are controlled by humans.
- Level 1: Certain systems, for example cruise control مثبت السرعة or automatic braking, can be controlled by the car.
- Level 2: At least two simultaneous automated functions such as acceleration and steering توجيه are offered by the vehicle. However, humans are required to ensure safe operation (emergency) .
- Level 3: The car is able to manage safety functions under certain conditions, but the driver is expected to take over when alerted.
- Level 4: The car is fully-autonomous in some driving scenarios, though not all.
- Level 5: The car is completely autonomous in every situation (2050).

## Under the bonnet

How a self-driving car works

Signals from **GPS (global positioning system)** satellites are combined with readings from tachometers, altimeters and gyroscopes to provide more accurate positioning than is possible with GPS alone

1

**Lidar (light detection and ranging)** sensors bounce pulses of light off the surroundings. These are analysed to identify lane markings and the edges of roads

**Radar sensor**

2

**Video cameras** detect traffic lights, read road signs, keep track of the position of other vehicles and look out for pedestrians and obstacles on the road

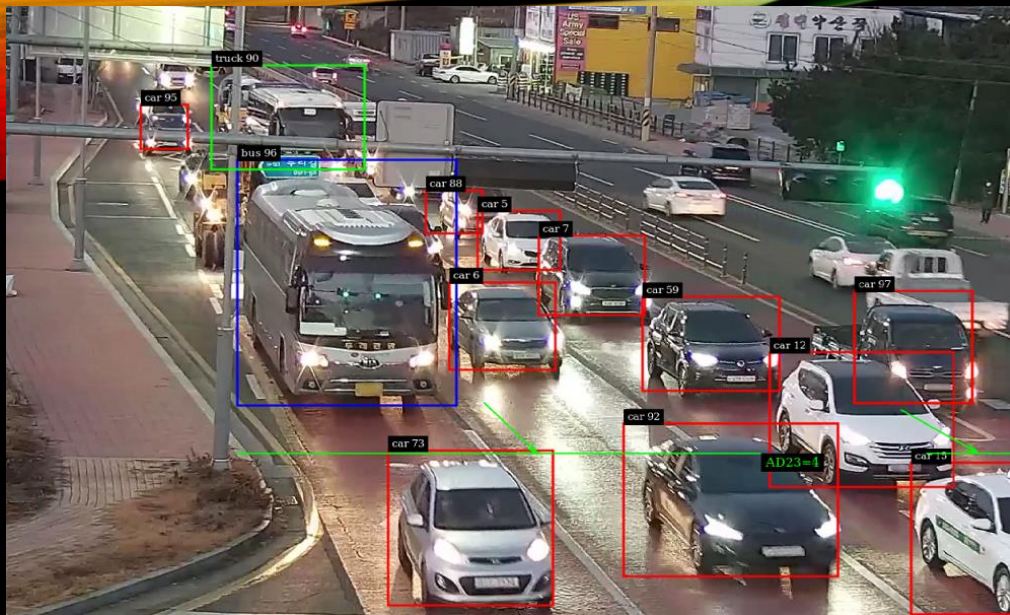
3

**Ultrasonic sensors** may be used to measure the position of objects very close to the vehicle, such as curbs and other vehicles when parking

The information from all of the sensors is analysed by a **central computer** that manipulates the steering, accelerator and brakes. Its software must understand the rules of the road, both formal and informal

4

**Radar sensors** monitor the position of other vehicles nearby. Such sensors are already used in adaptive cruise-control systems



**Normal  
Traffic on  
streets**



**monitored Traffic  
on centralized  
Operation  
Centers in Traffic  
Control systems**

# AI TRANSPORTATION (2050)

- There is the **Super Cruise** from General Motors (**GM**), which controls the vehicle on long highway stretches when not much is happening.
- There is the **Traffic Jam Assistant** from **BMW**, where cars move along in a jammed traffic area
- There's also **Road Train** where **one vehicle with a professional driver** leads a group of vehicles. These will be the bridge to **2050**, when we should be at level 6, which would mean **fully driverless cars**.
- **Drones** will fill the skies as urban transportation and floating warehouses deliver packages supported by **pick-up centers in population centers**, replacing much of the retail of today.

# WORLD POPULATION DISASTER

- The **current** world population of 7.6 billion is expected to reach:
  - 8.6 billion in 2030,
  - 9.8 billion in 2050 and
  - 11.2 billion in 2100, according to a new United Nations report being launched in 2023.

# WORLD POPULATION DATA

TOP WORLD POPULATION RANKINGS IN 2050 WILL STACK UP DIFFERENTLY THAN IN 2017

2017

|               |               |
|---------------|---------------|
| CHINA         | 1,387 MILLION |
| INDIA         | 1,353 MILLION |
| UNITED STATES | 325 MILLION   |
| INDONESIA     | 264 MILLION   |
| BRAZIL        | 208 MILLION   |
| PAKISTAN      | 199 MILLION   |
| NIGERIA       | 191 MILLION   |

2050

|               |               |
|---------------|---------------|
| INDIA         | 1,676 MILLION |
| CHINA         | 1,343 MILLION |
| NIGERIA       | 411 MILLION   |
| UNITED STATES | 397 MILLION   |
| INDONESIA     | 322 MILLION   |
| PAKISTAN      | 311 MILLION   |
| BRAZIL        | 231 MILLION   |

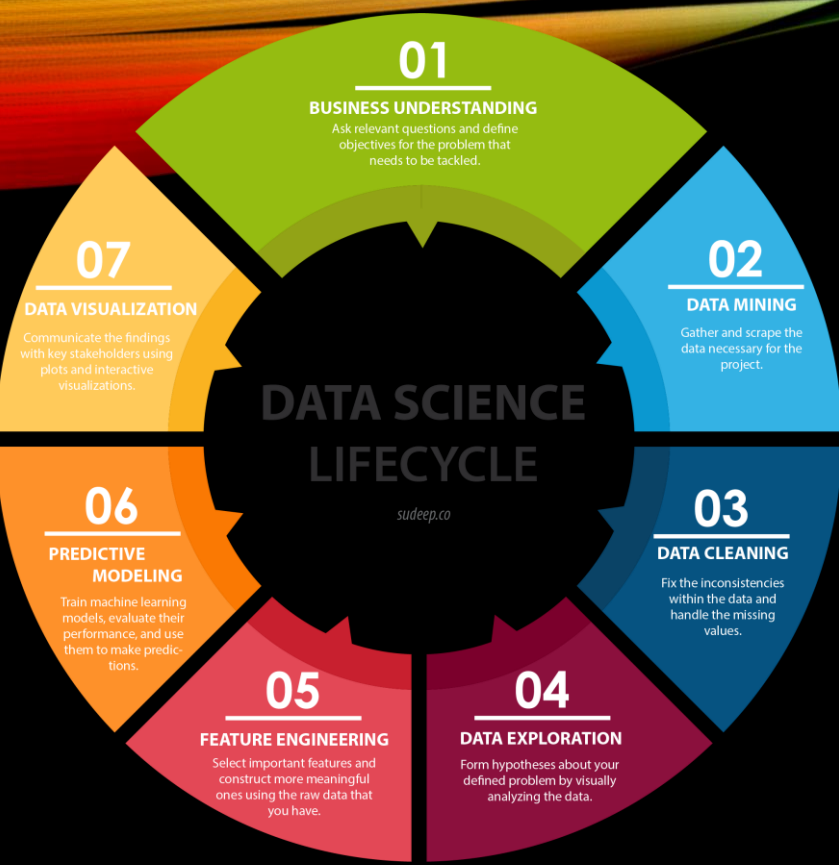
SOURCE: POPULATION REFERENCE BUREAU • WORLDPOPDATA.ORG • #WORLDPOPDATA



**Traffic with 9 lanes in delhi (India) in 2017.**  
**What about ambulance and pregnant women's?**  
**So, solving the problem of jam traffic is urgent.**

## 5. HEALTHCARE GOES GENOMIC AND PREDICTIVE

- Prevention is better than cure.
- We're moving from a system of **generalized healthcare** based on some of your high-level identifiers like age, race, and Body Mass Index (BMI).
- We're moving to a world of **personalized medicine based on your sequence genome**.
- the **genome** ( the entire set of DNA (Deoxyribonucleic acid ) instructions found in a cell data will be used to innovate the prevention via the predictive analytics aimed at making a **prediction of future events** based on the available current data.
- Once you, and millions of others for algorithm training, are sequenced, then healthcare **will be predictive**.
- This will enable early intervention, prevention, and personalized treatment. That will improve the chances of recovery and survival.
- AI will also assist doctors in making accurate diagnoses to **identify patients' conditions early on, and avoid the risk of complications**.



AI with Data Science will treat, and largely eliminate, neurological disorders like Alzheimer's, Parkinson's, most birth defects, and spinal cord الحبل الشوكي injuries as well as blindness and deafness.



By 2050, people will commonly **insert chips inside their bodies** and 'enhance' themselves, says Heather Delaney, founder of gallium ventures, a tech PR and comms consultancy in London.

The chips will monitor conditions within the body and enable people to **stay in touch with their health.**

# FEATURES OF HEALTH CARE CENTER (2050)

## The Rise of Personalized and Precision Medicine using inserted chips in bodies due to:

1. **Genomics** will enable us to create therapies that modify or replace defective genes with healthy ones.
2. **Proteomics** will help us understand how diseases affect proteins. How they can be used as biomarkers or targets for drug development.
3. **Metabolomics** can tell insights into disease mechanisms and potential interventions.

## The Expansion and Integration of Digital Technologies

1. Mobile Health (mHealth)
2. Telemedicine
3. Electronic health records (EHRs)
4. Blockchain allows secure and transparent transactions among multiple parties without intermediaries

## The Emergence of New Models and Roles of Care

1. Smart devices and sensors
2. Predictive analytics and AI
3. Nanobots and bioprinting
4. Remote care and VR

## 6. WORKPLACE TRANSFORMATION

- Administrative, sales, food service, transportation, and manufacturing jobs, will see massive disruption in how we work .
- New work philosophy will emerge (**on line jobs**).
- **General AI**, where machines (**Robotics**) have the capacity to understand or learn any intelligent task that a human being can.
- **Neural networks** were built with a specific task in mind and trained on selected datasets are increasingly necessary.
- **AI can find the right data, make the calls, and execute.**

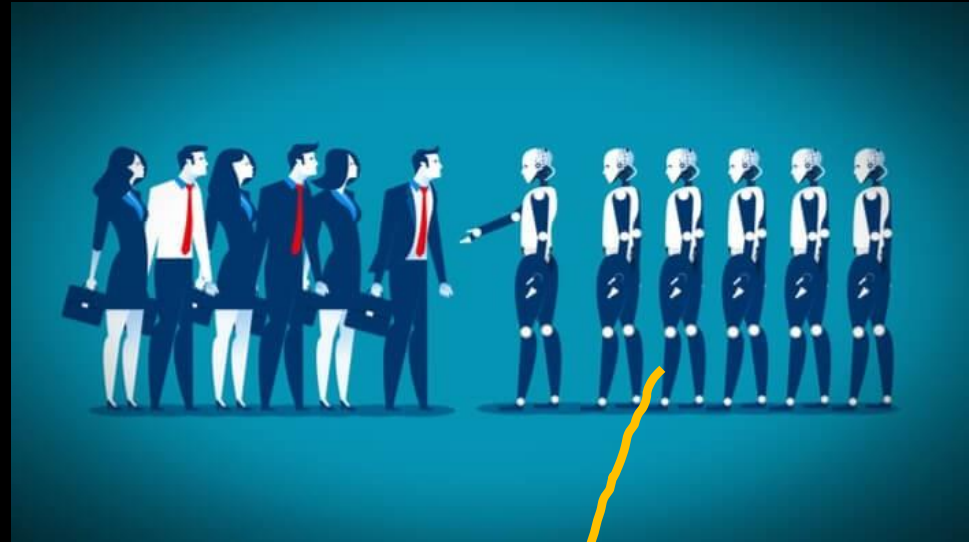
# JOB MARKET (2050)

- Job market will differ as follows:

- Robots have rights too
- Personal assistants for life
- A flexible workforce
- Focus on the user experience
- Automation AI, and emerging industries will **create new opportunities while transforming traditional careers.**
- **Soft skills** such as emotional intelligence, critical thinking, and problem-solving will be more important than ever before.
- Those who are willing to **adapt and learn** will be the ones who **succeed** in the job market of the future.
- So, it's important to **start developing new skills, stay ahead of trends, and**  
**be ready for the changes ahead.**

# 10 JOBS THAT WILL DISAPPEAR (2050)

1. Taxi Drivers
2. Printing Press Operators
3. Textile Worker
4. Postal worker
5. Fast Food Worker
6. Cashier
7. Travel Agent
8. Switchboard Operators
9. Customer Support
10. Warehouse Worker



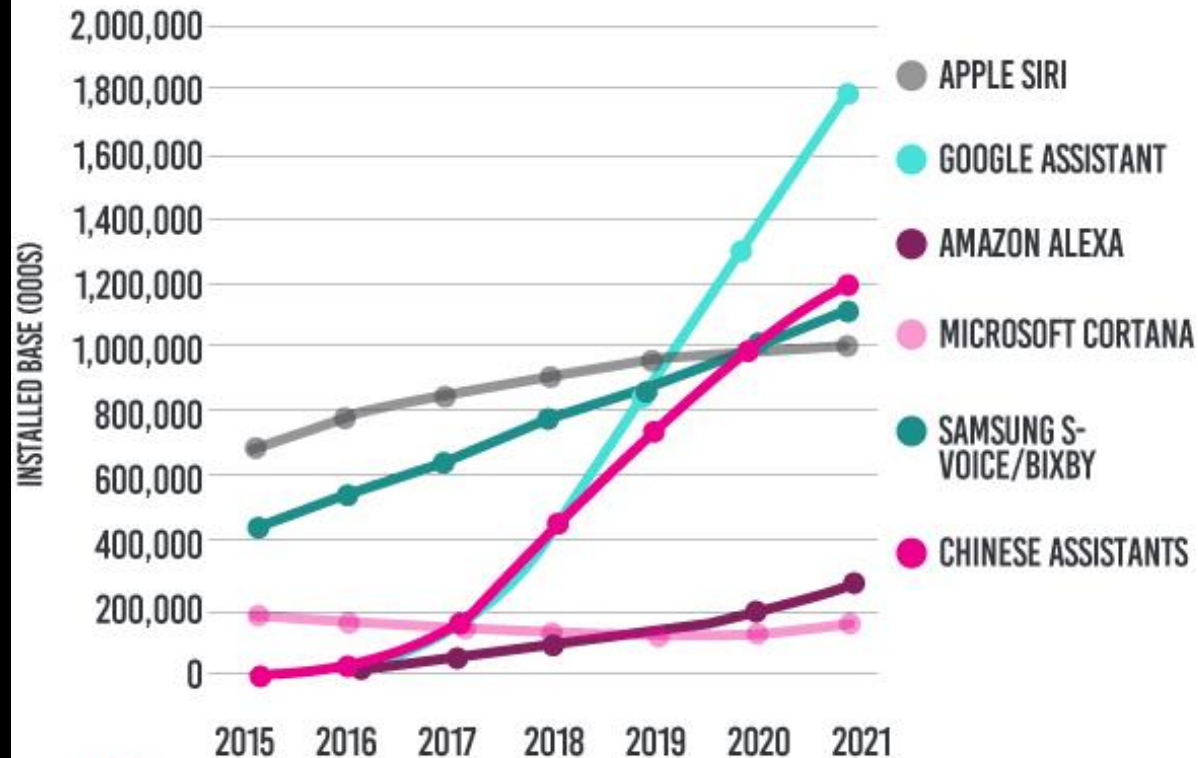
Robotics ( AI coworkers)

# 7. AI IN SPACE EXPLORATION (2050)



# AI ASSISTANTS' MOMENTUM

MORE THAN **7.5 BILLION** ACTIVATED DIGITAL ASSISTANTS DEVICES BY 2021



50%

AND MORE OF MILLENNIALS RELY ON DIGITAL ASSISTANTS BY MAKING VOICE COMMANDS **AT LEAST ONCE A MONTH.**

Personal AI Assistant

Is the digital assistant a friend or a spy???????

## 9. EDUCATION (2050)

- Interactive and **mix Learning** with AR, VR, and AI
- **3D Printers**: From Imagination to Reality
- **Cognitive Learning** Enhanced by AI
- **Instant** Feedback and Smart Attendance
- Universal Internet Accessibility and the **Rise of Remote Learning**
- The **Metaverse** as a Learning Platform
- Literature Reimagined with AR

# CLASSROOM (2050)



# 10. AI IN SPORT AND FITNESS

- No need for personal trainer, AI coaches could provide personalized training programs, while AI systems could manage sports leagues, optimize team strategies, and even predict injuries before they occur.
- AI-driven wearables, fitness trackers, and smartwatches have become helpful assistants for monitoring physical activity and sports.
- Brands like NIKE will construct a special video advertisement for somebody that likes to take long walks with comfortable shoes and will construct another specialized advertisement for the same shoe for another person who likes to play football, i.e. the influencers will use AI in content creation on social media platforms that is more tailored to individual audiences, resulting in higher engagement

# OUR DAILY LIFE(2050)

## Morning - AI Personal Assistants and Smart Homes

- **Wake-Up Routine:** The day begins with an **AI personal assistant** gently waking up **Sarah**, our hero, using a light and sound sequence optimized for her sleep cycle. The room's temperature and lighting adjust to her preferences, ensuring a comfortable day start.
- **Smart Wardrobe:** As Sarah gets ready, her **AI clothes** suggests an outfit based on the day's weather forecast and her schedule. The clothes are made of smart fabrics that adapt to temperature changes.
- **Nutrition and Health:** In the kitchen, her **AI nutritionist prepares a breakfast** customized to her eating needs, using ingredients from her smart fridge. Her **health monitoring system** updates her on her vitals and fitness progress.

# OUR DAILY LIFE (2050)

## Travel - Autonomous Vehicles and AI in Transit

- **Smart Travelling:** Sarah steps into her self-driving car, which communicates with city traffic AI to find the quickest, most efficient route. During the travel, she uses this time to catch up on news, selected by her AI assistant based on her interests.
- **Environmental Impact:** The car is electric and part of a shared vehicle network, significantly reducing carbon footprints and traffic crowding

## Work - AI Collaboration and Enhanced Productivity

- **AI Coworkers:** At work, Sarah collaborates with AI systems that help with data analysis, creativity, and problem-solving, making her job more efficient and allowing her to focus on complex tasks.
- **Virtual Meetings:** Advanced virtual reality meetings allow real-time collaboration with colleagues worldwide, with AI translation removing any language barriers.

# OUR DAILY LIFE (2050)

## Afternoon - AI in Education and Leisure

- **Learning and Development:** Sarah attends an AI-powered education session, offering a personalized learning experience. The AI adapts the curriculum to her learning style and pace.
- **Leisure Activities:** For relaxation, she visits a virtual reality park, experiencing a safari adventure created by AI to simulate real-world experiences.

## Evening - Social and Family Life

- **Home Cooking:** Back home, an AI-powered kitchen helps Sarah cook a meal, offering step-by-step instructions and adjusting recipes to her taste.
- **Family Time:** Sarah enjoys family time with interactive AI-powered games and educational tools, offering a fun and engaging way for her children to learn and play.

## Night - Health and Comfort

- **Wellness Check:** Before bed, her health AI conducts a quick wellness check, adjusting her sleep environment to ensure a restful night.
- **Reflection and Planning:** Sarah reviews her day and plans for tomorrow with her AI assistant, which organizes her schedule and sets reminders for important tasks.

# Terrifying AI

## ( Robotics Self Independence)

- In June 2023 , DeepMind released Robocat as a self-improving robotic agent which is an update on last year's Gato),
- It generates its own data from trial and error to learn how to control many different robot arms to do multitasks from the observation of 100 demonstrations
- So, it reduces the need for human-supervised training, and is an important step towards creating a general-purpose robot with their flexibility and low cost



Robocat

# Crazy AI Humans Integration

- Imagine a world where:
  - **artists** collaborate with AI thinks,
  - **scientists** brainstorm with silicon super minds,
  - **engineers** design alongside virtual partners who **can predict material properties and test prototypes in milliseconds.**
- Beyond cognitive enhancement, AI integration could revolutionize **healthcare.**
- **Brain-computer interfaces** could treat neurological disorders, restore lost senses, with unparalleled precision. **Imagine:**
  - paralyzed individuals walking again,
  - deaf hearing symphony orchestras, and
  - blind painting masterpieces,

**All guided by the invisible hand of AI.**

# Could Half Our Brain Be An AI?

- **Imagine if half of your brain wasn't yours, but an artificial intelligence** — a living, silicon partner constantly analyzing, optimizing, and whispering suggestions in your ear.
- It sounds like science fiction, but the concept of integrating AI directly into the human brain is no longer referred to movies like "Ghost in the Shell 2017."
- With advancements in neural interfaces and AI development, the idea of a "cyborg brain" is closer to reality than you might think.



# AI Ethical Frameworks

## ( next speech topic)

- As AI becomes more deeply embedded in our lives, the need for robust ethical frameworks will be essential.
- **The big questions are :**
  - would we lose the ability to make our own decisions?
  - What about our Data Privacy and Security?
- 2050 will witness a great effort to ensure that AI is developed:
  - responsibly,
  - protection against:
    1. bias
    2. discrimination
    3. unintended consequences

